SURF Summer Undergraduate Research Fellowship

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FX Risk Sentinel: AI and Econometrics for Predictive Risk Analysis

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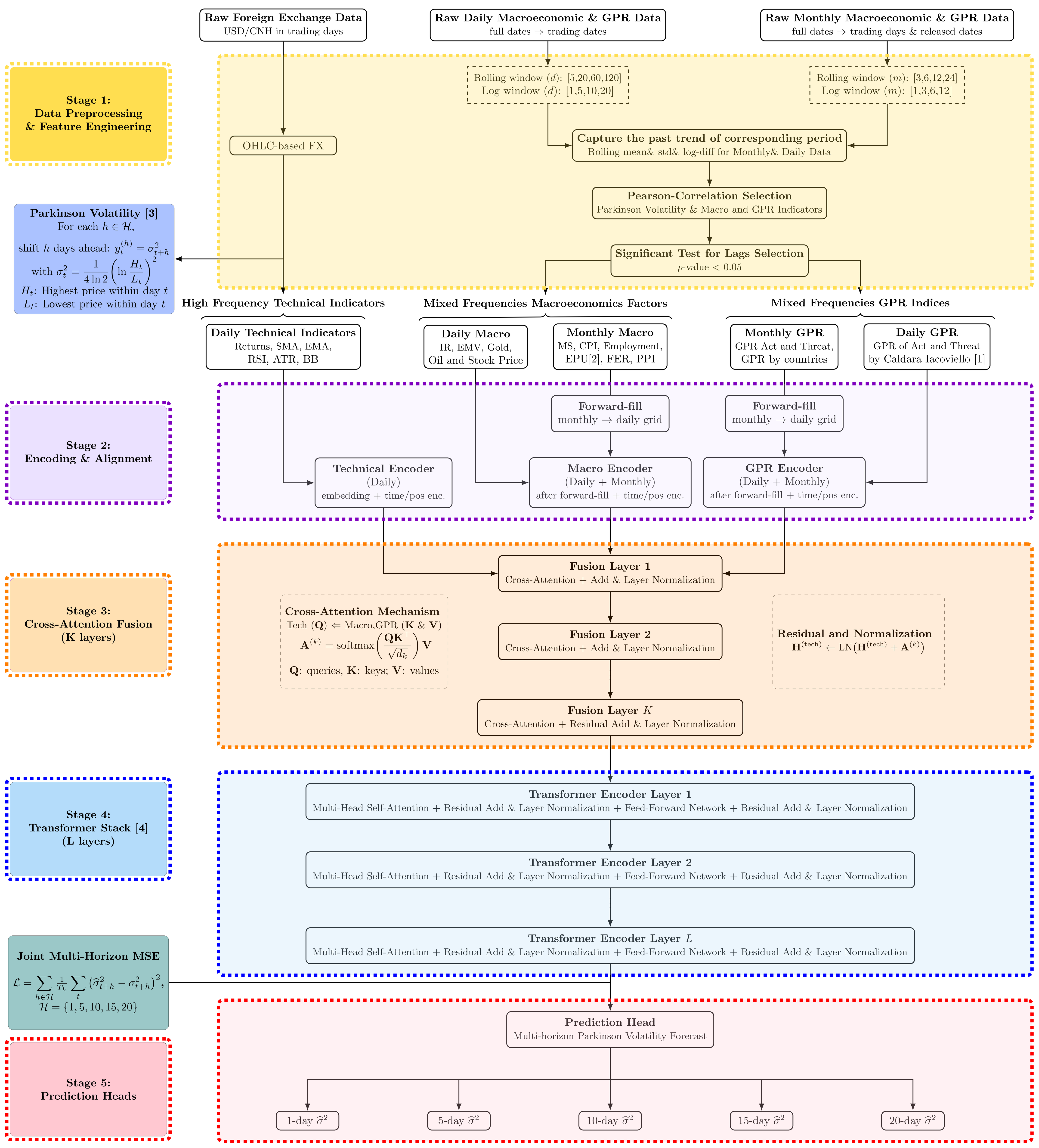
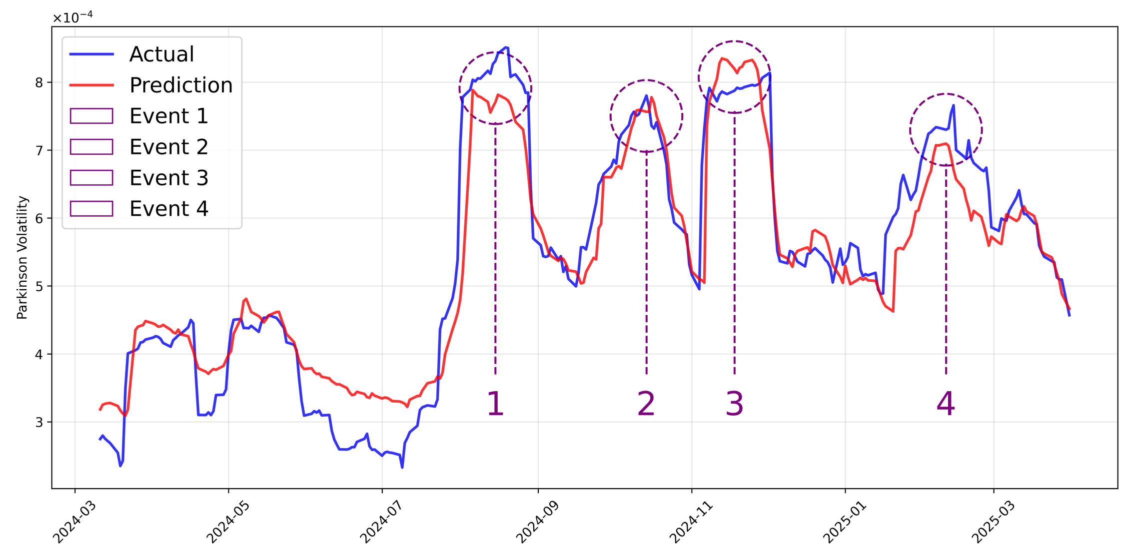
Abstract

Our interdisciplinary team, with members from the School of Mathematics and Physics, the School of Advanced Technology, and the International Business School Suzhou, developed a robust FX risk signal detection system. By integrating technical, macroeconomic, and geopolitical risk indicators, we designed a mixed-frequency transformer-based model, deployed through both a web platform and an iOS application. These platforms generate daily risk signals, issue real-time alerts, and adjust hedging strategies to enhance returns while mitigating risk. The system has been validated by successfully capturing FX risk spikes between March 2024 and March 2025, demonstrating its reliability as a tool for crisis resilience.

Methodology



Results



USD/CNH Volatility: Mar. 2024 – Mar. 2025

- **Event 1 (Aug. 2024)**: Record VIX shock & carry trade unwind
- **Event 2 (Oct. 2024)**: Tariff tensions & weak Chinese stimulus
- **Event 3 (Nov. 2024)**: U.S. presidential election shock
- **Event 4 (Feb. 2025)**: Realized U.S. tariff implementation

Metric	Description	Value
RMSE	Precision	5.41×10^{-5}
MAE	Consistency	4.07×10^{-5}
MAPE	Relative Accuracy	9.01%
DA	Directional Prediction	63.64%
R^2	Explanatory Power	90.23%

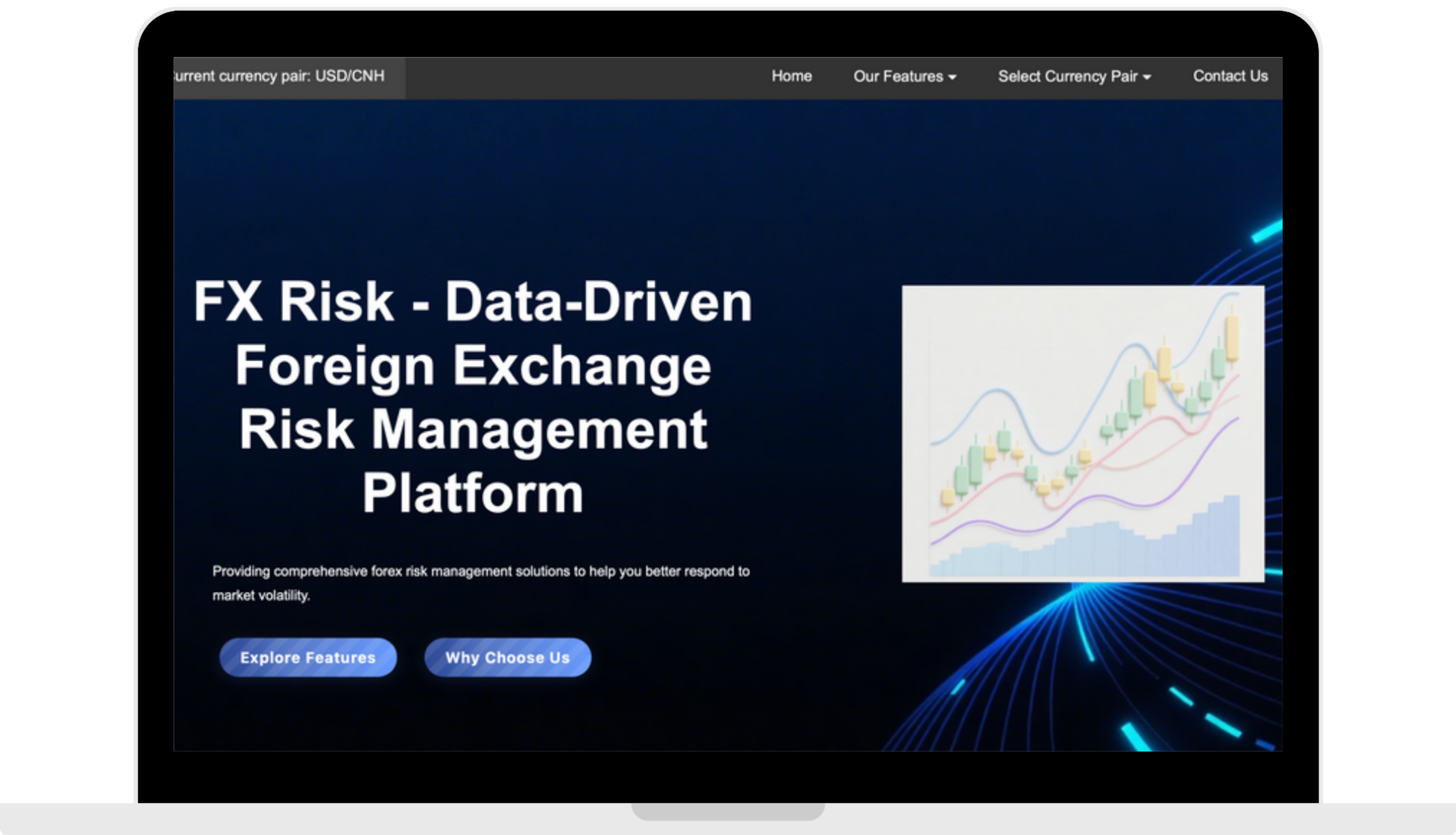
Model Accuracy & Risk Detection Evaluation

Conclusion

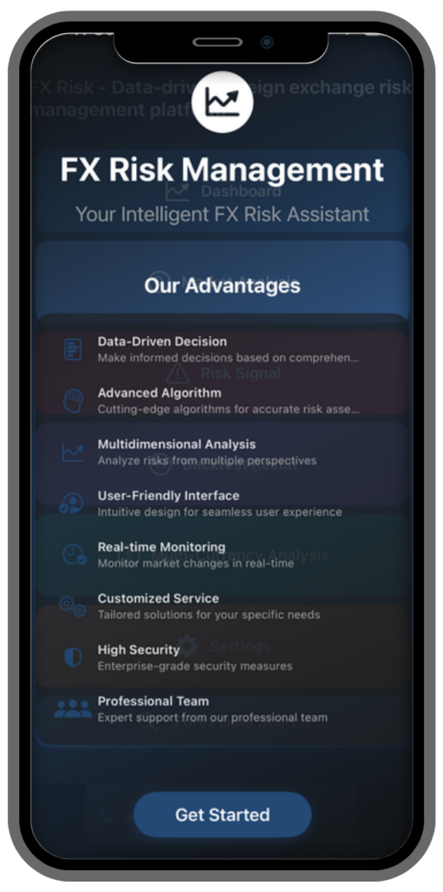
Our model based on the 20-day rolling window with 1-day prediction horizon demonstrates robust predictive performance for USD/CNH Parkinson volatility and achieves strong precision accuracy (RMSE = 5.41×10^{-5}), high explanatory power ($R^2 =$ 90.23%), and high directional accuracy (DA= 63.64%).

Beyond this, our framework exhibits generalization capability across short to medium term horizons (1–10 days), which effectively captures daily FX risk signals. Notably, macroeconomic indicators play a crucial role in enhancing medium-to-long-term predictions, while daily GPR indices significantly contribute to detecting short-term risk spikes especially for geopolitical shocks. This stresses the system's adaptability in both crisis anticipation and strategic hedging.

Product Development



Please scan for the website demo video.



Please scan for the iOS demo video.

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